Iván Martín Hernández

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Professional Profile

Computational scientist with a PhD in Biochemistry, Molecular Biology and Biomedicine, specialised in Structural Bioinformatics. Experienced in the development of computational methods and pipelines for biological research, with a strong emphasis on scientific programming, algorithm design, and data-driven analysis. Background in protein structure complemented by experience in sequence and genomic-based studies and reproducible computational workflows.

Employment History

2026 (3 months) — **Postdoctoral Researcher.** I performed data analysis on a cohort of patients with Autism Spectrum Disorder (ASD) at the Health Research Institute of Hospital General Universitario Gregorio Marañón, within a personalized medicine project.

2020 – 2025 — **PhD researcher.** I completed my doctoral thesis in the structural bioinformatics laboratory of Pablo Chacón, at the Institute of Chemistry and Physics Blas Cabrera (IQF-CSIC). I developed a specific version of the algorithm Random Coordinate Descent (RCD), protein loop structure modeling, specialized in predicting antibody CDRs (AbRaCD).

2019 – 2020 — **Support researcher at the Carlos III University (UC3M).** Participation in two European FAIRification projects of scientific data: FAIRsFAIR and FAIR4Health.

2014 – 2019 — **Research Internships & Early Experience.**

- **2019** (7 months): **MSc Research Intern** in the Structural Bioinformatics group of the IQF-CSIC, focused on protein loop structure prediction.
- **2018** (6 months): **iGEM Team Member** (Madrid-OLM), computational aptamer modelling and project wiki development (*"Internet of Biothings"*, Silver Medal).
- **2017** (6 months): **BSc Research Intern** in the Virology Group at the Biotechnology and Plant Genomics Center (CBGP-UPM), combining wet-lab and computational work.
- **2015** (1 month): **Summer Research Intern** in the Bioinformatics group at CBGP-UPM.
- **2014** (2 months): **Science Outreach Collaborator** with the AsBioMad asociation.

Education

2020 – 2025 — **Ph.D.** in **Biochemistry, Molecular Biology and Biomedicine** at the Complutense University of Madrid (UCM). Majoring in **Structural Bioinformatics**.
Thesis title: *Desarrollo y aplicación de herramientas bioinformáticas para la predicción de estructura de bucles de proteínas.* 'Development and application of bioinformatics tools for protein loop structure prediction' (Text in Spanish).
– Excellent "**Cum Laude**".

2018 – 2019 — **M.Sc.** in **Computational Biology** at the Polytechnic University of Madrid (UPM).
Master thesis title: *Development and application of bioinformatics tools for protein loop modelling*.

2013 – 2017 — **B.Sc.** in **Biotechnology** at the Polytechnic University of Madrid (UPM). Majoring in **Bioinformatics**.
Final degree work: *El solapamiento de genes como modulador de la velocidad de evolución en virus de ADN de plantas*. 'Gene overlaping as a modulator of evolution speed in plant DNA virus' (Text in Spanish).

Research Publications

Journal Articles

1. **I. Martín-Hernández**, P. Chacón, and J. R. López-Blanco, "Abracd: Fast sampling-based prediction of antibody cdr loops," *Bioinformatics Advances (Submitted)*, 2026.
2. **I. M. Hernández**, Y. Dehouck, U. Bastolla, J. R. López-Blanco, and P. Chacón, "Predicting protein stability changes upon mutation using a simple orientational potential," *Bioinformatics*, vol. 39, no. 1, A. Valencia, Ed., Jan. 2023. DOI: 10.1093/bioinformatics/btad011
3. **I. Martín-Hernández** and I. Pagán, "Gene overlapping as a modulator of begomovirus evolution," *Microorganisms*, vol. 10, no. 2, p. 366, Feb. 2022. DOI: 10.3390/microorganisms10020366
4. J. Beal et al., "Robust estimation of bacterial cell count from optical density," *Communications Biology*, vol. 3, no. 1, 2020. DOI: 10.1038/s42003-020-01127-5
5. M. Grootveld et al., "Workshop report "training in the eosc"," 2020. DOI: 10.5281/ZENODO.3894370

Conference Proceedings and Posters

1. **I. M. Hernández**, J. R. Lopéz-Blanco, and P. Chacón, "Improving protein loop modelling for antibody h3-cdr loops.," in *V PhDay Biología UCM*, 2021.
2. **I. Martín-Hernández**, J. R. López-Blanco, and P. Chacón, "Improving protein loop modelling for antibody h3-cdr loops.," in *7th International Iberian Biophysics Congress - CongressBook*, 2021, ISBN: 978-989-33-2042-6.
3. **I. Martín-Hernández**, J. R. López-Blanco, and P. Chacón, "Improving loop modeling prediction," eng, in *JOINT 12th EBSA congress and 10th ICBP – IUPAP congress, July 20-24, 2019, Madrid, Spain – Abstracts*, European Biophysics Journal, 2019. DOI: https://doi.org/10.1007/s00249-019-01373-4
4. Madrid-OLM, "Internet of biothings," in *iGEM 2018 Giant Jamboree*, Boston, Oct. 2018.

Tools and Servers

AbRaCD **Antibody Random Coordinate Descent**.
Antibody CDR loop structure prediction server.
– https://abracd.iqf.csic.es

Skills

Languages	Native **Spanish**. Professional **English**.
Programming	Python, Perl, R, C++, Bash.
Computing	Linux/Unix, HPC clusters, parallel computing, Git.
Structure	Protein modelling, Molecular Dynamics, Rosetta, AlphaFold, PyMOL, Chimera, VMD, AutoDock, RCD, AbRaCD, KORP, KORPM, Mutation stability prediction.
Phylogeny	MEGA, Clustal, Mesquite, HyPhy, FigTree, BEAST & BEAUti, PhyML.
Genomics	NGS, variant calling, RNA-seq, ChIP-seq, biomarker identification, genome assembly and comparative genomics.
Data Science	Machine learning analysis, SQL, Weka, Protégé.
Web	Basic HTML and CSS.

References

Dr. Pablo Chacón Montes, PhD
– Research Scientist (CSIC) and Head of the Structural Bioinformatics Group.
– Institute of Chemistry and Physics Blas Cabrera (IQF-CSIC), Madrid, Spain.
– *PhD Supervisor (2020–2025).*
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Dr. José Ramón López-Blanco, PhD
– Postdoctoral Researcher, Structural Bioinformatics Group.
– Institute of Chemistry and Physics Blas Cabrera (IQF-CSIC), Madrid, Spain.
– *Senior Researcher (2020–2025).*
jrlopez@iqf.csic.es

Prof. María José Parellada Redondo, MD, PhD
– Professor of Psychiatry, Complutense University of Madrid (UCM).
– PI at the Health Research Institute of Hospital General Universitario Gregorio Marañón, Madrid, Spain.
– *Research Supervisor (2026).*
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